Lesson 5.3 - Probability of Complementary Events

> **Video:** Probability of Complementary Events & Sample Space by The Organic Chemistry Tutor **Chapter:** 5 - Introduction to Probability

Setup - rolling one six-sided die

- Sample space $S = \{1, 2, 3, 4, 5, 6\}$
- A = numbers **at most 4** = $\{1, 2, 3, 4\}$
- $B = \{3, 4, 5\}$

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1. $P(A \text{ and } B)$ - intersection

$A \cap B = \{3, 4\}$ (the outcomes in **both** sets)

$$P(A \text{ and } B) = \frac{n(A \cap B)}{n(S)} = \frac{2}{6} = \frac{1}{3} \approx 0.33$$

2. $P(A \text{ or } B)$ - union

$A \cup B = \{1, 2, 3, 4, 5\}$ (the outcomes in **either** set; only 6 is missing)

$$P(A \text{ or } B) = \frac{n(A \cup B)}{n(S)} = \frac{5}{6} \approx 0.833 = 83.3\%$$

3. $P(A')$ - complement

The **complement** A' is everything in S that is **not** in A: $A' = \{5, 6\}$

$$P(A') = \frac{n(A')}{n(S)} = \frac{2}{6} = \frac{1}{3}$$

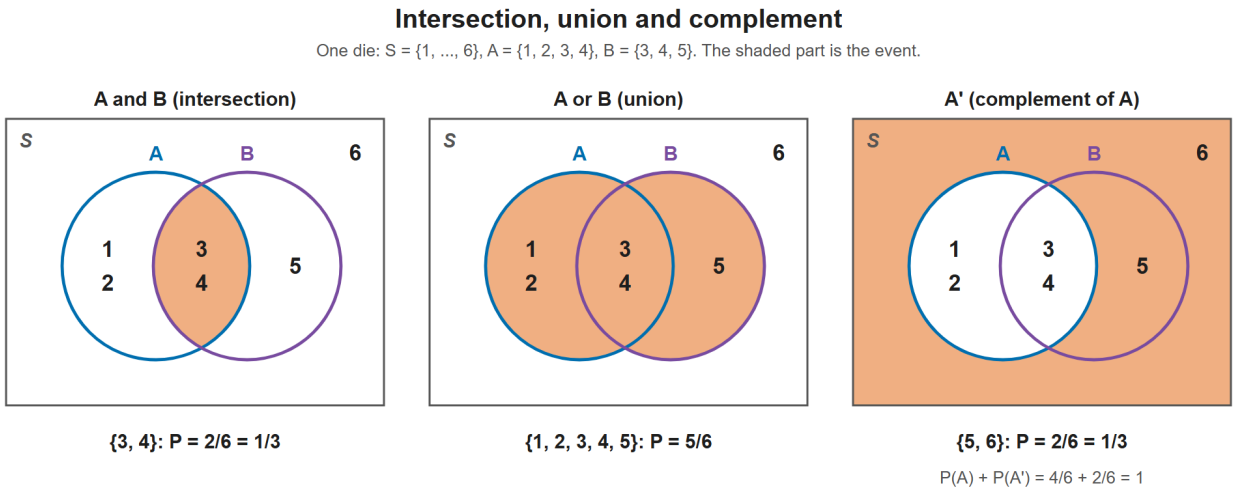
4. $P(A)$ and the complement rule

$$P(A) = \frac{n(A)}{n(S)} = \frac{4}{6} = \frac{2}{3}$$

$$P(A) + P(A') = \frac{2}{3} + \frac{1}{3} = 1$$

An event and its complement together cover the whole sample space, so their probabilities add to 1:

$$\boxed{P(A') = 1 - P(A)}$$



The same two events shaded three ways. The complement A' is everything in S outside A, so A and A' together fill the whole box: P(A) + P(A') = 1.

Example - marbles

The probability of drawing a **red** marble from a bag is 35%. What is the probability of drawing a marble that is **not red**?

$$P(\text{not red}) = 1 - P(\text{red}) = 1 - 0.35 = 0.65 = 65\%$$

Key takeaways

- Intersection (A and B) = outcomes in **both**. Union (A or B) = outcomes in **at least one**.
- Complement A' = outcomes **not** in A.
- $P(A) + P(A') = 1$, so $P(A') = 1 - P(A)$. This is often the fastest route, especially for "at least one" problems.